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Smart Internz

Model Development Phase Template

Date	28 June 2025
Student ID	SWUID20250181505
Project Title	online payments fraud detection using machine learning
Maximum Marks	6 Marks

Model Selection Report

In the forthcoming Model Selection Report, various models will be outlined, detailing their descriptions, hyperparameters, and performance metrics, including Accuracy or F1 Score. This comprehensive report will provide insights into the chosen models and their effectiveness.

Model Selection Report:

Model	Description	Hyperparameters	Performance Metric (e.g., Accuracy, F1 Score)
Random Forest Classifier	An ensemble method that builds multiple decision trees and aggregates their predictions, robust to overfitting and effective for imbalanced datasets after preprocessing. Selected for deployment due	Default settings	Test Accuracy: 0.9733 (97.33%), F1-Score (Fraud): 0.97, F1-Score (Non-Fraud): 0.98

	to high accuracy and interpretability.		
Decision Tree Classifier	A single decision tree that splits data based on feature thresholds, is simple but prone to overfitting, especially on imbalanced data.	Default settings	Test Accuracy: 0.9383 (93.83%), F1-Score (Fraud): 0.93, F1-Score (Non-Fraud): 0.94
Extra Trees Classifier	An ensemble method similar to Random Forest but with randomized splits, reducing variance but potentially less interpretable.	Default settings	Test Accuracy: 0.9650 (96.50%), F1-Score (Fraud): 0.96, F1-Score (Non-Fraud): 0.97
Support Vector Classifier (SVC)	A classifier that finds the optimal hyperplane to separate classes, effective in high-dimensional spaces but	Default settings	Test Accuracy: 0.7695 (76.95%), F1-Score (Fraud): 0.66, F1-Score (Non-Fraud): 0.82

	computationally intensive and less effective for this dataset.		
XGBoost Classifier	A gradient-boosting algorithm that builds sequential trees to minimise errors is highly effective for complex patterns and imbalanced data.	Default settings	Test Accuracy: 0.9835 (98.35%), F1-Score (Fraud): 0.98, F1-Score (Non-Fraud): 0.98